The pigeonhole principle

High school · olympiad thinking · Worked solutions · 02

Exercise 1. Show that among 25 people, at least three were born in the same month.

Solution. At most two per month would give at most 24 people. Therefore one month contains at least three.

Exercise 2. Choose six distinct numbers from $\{1, \dots, 10\}$. Prove that two sum to 11.

Solution. The five boxes are $\{1,10\}$, $\{2,9\}$, $\{3,8\}$, $\{4,7\}$, $\{5,6\}$. Six choices fill at least one pair.

Exercise 3. Prove that among eight integers, two have a difference divisible by 7.

Solution. There are seven possible remainders modulo 7. Two integers share a remainder, so their difference is a multiple of 7.

Exercise 4. Five points lie in a square of side 2, including its boundary. Show that two are at distance at most $\sqrt{2}$.

Solution. Divide the square into four unit squares, assigning each boundary point to just one square. Two points share a unit square, whose diagonal is $\sqrt{2}$.

Problem · Investigate, justify, conclude. Choose 11 distinct integers from 1 to 20. Write each as $2^k m$ with m odd and $k \geq 0$. How many values of m are possible? Prove that one selected integer divides another. Show that the conclusion can fail with only ten integers.

Solution. Remove all factors of 2. The ten possible odd parts are $1, 3, \dots, 19$. Two chosen numbers share an odd part: $2^a m$ and $2^b m$, with $a < b$ because they are distinct. The second is 2^{b-a} times the first. For ten numbers choose $11, 12, \dots, 20$: twice any of them exceeds 20, so none divides another distinct one.